

MSTAR PRACTICE WORKSHEET**Lesson 6-1: Division Expressions with Fractions and Whole Numbers**

Grade 6 Mathematics · Standard 6.NOS.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: AWhat does $3 \div \frac{1}{4}$ mean?

- ☒ A How many $\frac{1}{4}$ -size pieces fit into 3
- ☐ B $\frac{1}{4}$ of 3
- ☐ C 3 groups of $\frac{1}{4}$
- ☐ D 3 minus $\frac{1}{4}$

$3 \div \frac{1}{4}$ asks how many $\frac{1}{4}$ -size pieces fit into 3. To find that, put a 1 under the 3 and use Keep, Change, Flip: $3/1 \times 4/1 = 12$.

Question 2 SELECTED RESPONSE — CORRECT: BWhich expression means 'how many $\frac{1}{3}$ -size pieces are in 2'?

- ☐ A $\frac{1}{3} \div 2$
- ☒ B $2 \div \frac{1}{3}$
- ☐ C $2 \times \frac{1}{3}$
- ☐ D $2 + \frac{1}{3}$

Finding how many $\frac{1}{3}$ -size pieces fit into 2 is written $2 \div \frac{1}{3}$. Put a 1 under the 2, then Keep, Change, Flip: $2/1 \times 3/1 = 6$.

Question 3 **SELECTED RESPONSE — CORRECT: D**

A rope is 4 feet long. How many $\frac{1}{2}$ -foot pieces can be cut from it?

- ☐ A $\frac{1}{2}$ piece
- ☐ B 2 pieces
- ☐ C 4 pieces
- ☒ D 8 pieces

Write $4 \div \frac{1}{2}$. Put a 1 under the 4, then Keep, Change, Flip: $4/1 \times 2/1 = 8$ pieces.

Question 4 **SELECTED RESPONSE — CORRECT: A**

A half-pan of brownies is shared equally among 4 people. How much of the whole pan does each get?

- ☒ A $\frac{1}{8}$
- ☐ B $\frac{1}{2}$
- ☐ C 2
- ☐ D $\frac{4}{2}$

This is $\frac{1}{2} \div 4$. The whole number is second this time, so put a 1 under the 4, then Keep, Change, Flip: $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$ of the whole pan.

Question 5 **SELECTED RESPONSE — CORRECT: C**

Agent Cole has $\frac{3}{4}$ of a pound of fingerprint powder and splits it equally into 2 kits. How much powder goes in each kit?

- ☐ A $1 \frac{1}{2}$ pounds
- ☐ B $\frac{1}{8}$ pound
- ☒ C $\frac{3}{8}$ pound
- ☐ D $\frac{3}{4}$ pound

Write $\frac{3}{4} \div 2$ as $\frac{3}{4} \div \frac{2}{1}$. Keep, Change, Flip: $\frac{3}{4} \times \frac{1}{2} = \frac{3}{8}$ pound in each kit. The top number does not have to be a 1 — the steps are exactly the same.

Question 6

SELECTED RESPONSE — CORRECT: A

What is $\frac{2}{3} \div 4$?

- ☒ A $\frac{1}{6}$
- ☐ B $\frac{2}{7}$
- ☐ C $\frac{3}{8}$
- ☐ D $\frac{8}{3}$

Write the 4 as $\frac{4}{1}$. Keep, Change, Flip: $\frac{2}{3} \times \frac{1}{4} = \frac{2}{12} = \frac{1}{6}$. Dividing by a whole number bigger than 1 makes the piece smaller, and $\frac{1}{6}$ is smaller than $\frac{2}{3}$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7**TWO-PART QUESTION****Part A — correct answer: D**

What is the value of $6 \div \frac{1}{5}$?

- ☐ A $\frac{1}{30}$
- ☐ B $\frac{6}{5}$
- ☐ C 11
- ☒ D 30

Put a 1 under the whole number so it becomes $\frac{6}{1}$. Keep $\frac{6}{1}$, change $\div$ to $\times$, and flip $\frac{1}{5}$ to its reciprocal $\frac{5}{1}$.

Multiply across: $\frac{6}{1} \times \frac{5}{1} = \frac{30}{1} = 30$.

- **A:** That is the reciprocal of the answer. Keep the 6 on top: $\frac{6}{1}$ times $\frac{5}{1}$.
- **B:** That multiplies 6 by $\frac{1}{5}$. Division flips the divisor first: $\frac{6}{1}$ times $\frac{5}{1}$.
- **C:** That adds 6 and 5. Keep, change, flip, and then multiply.

Part B — correct answer: B

Which reasoning shows why your answer to Part A is correct?

- ☐ A Keep $\frac{1}{5}$, change $\div$ to $\times$, and flip $\frac{6}{1}$ to $\frac{1}{6}$: $\frac{1}{5} \times \frac{1}{6} = \frac{1}{30}$.
- ☒ B Write 6 as $\frac{6}{1}$. Keep $\frac{6}{1}$, change $\div$ to $\times$, flip $\frac{1}{5}$ to $\frac{5}{1}$, then multiply across: $\frac{6}{1} \times \frac{5}{1} = 30$.
- ☐ C Write 6 as $\frac{6}{1}$ and multiply straight across without flipping: $\frac{6}{1} \times \frac{1}{5} = \frac{6}{5}$.
- ☐ D Add the whole number and the denominator of the fraction: $6 + 5 = 11$.

Keep, change, flip works only after the whole number is written as a fraction over 1, and only the second fraction gets flipped. That gives $\frac{6}{1} \times \frac{5}{1} = 30$.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about the expression $4 \div \frac{1}{3}$.

- ☒ **A** The whole number 4 can be written as the fraction $\frac{4}{1}$.
- ☒ **B** Keep, change, flip rewrites the expression as $\frac{4}{1} \times \frac{3}{1}$.
- ☒ **C** The reciprocal of $\frac{1}{3}$ is $\frac{3}{1}$.
- ☐ **D** The quotient is $\frac{4}{3}$.
- ☒ **E** The quotient is 12.
- ☐ **F** The quotient must be less than 4 because the expression uses division.

Write 4 as $\frac{4}{1}$, keep it, change $\div$ to $\times$, and flip $\frac{1}{3}$ to $\frac{3}{1}$. Then $\frac{4}{1} \times \frac{3}{1} = \frac{12}{1} = 12$. The quotient $\frac{4}{3}$ comes from multiplying by $\frac{1}{3}$ instead of dividing, and dividing by a number smaller than 1 makes the quotient larger, not smaller.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Dividing 6 by $\frac{1}{2}$ gives 12, but dividing 6 by 2 gives 3. How can dividing by a smaller number ($\frac{1}{2}$) give a bigger answer than dividing by a larger number (2)? Explain using a real-world example.

Model answer: The two expressions ask different questions, and the flip shows why. $6 \div 2$ is $\frac{6}{1} \times \frac{1}{2} = 3$, because flipping $\frac{2}{1}$ gives a factor smaller than 1. $6 \div \frac{1}{2}$ is $\frac{6}{1} \times \frac{2}{1} = 12$, because flipping $\frac{1}{2}$ gives a factor larger than 1. Dividing by a number less than 1 always makes the quotient grow.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.